

Shaurya Gomber

Education

- 2024–Present **Ph.D. in Computer Science**, *University of Illinois Urbana-Champaign*, USA
GPA: 4.0/4.0; Advisor: [Prof. Gagandeep Singh](#)
- 2022–2024 **M.S. in Computer Science**, *University of Illinois Urbana-Champaign*, USA
GPA: 4.0/4.0; Advisor: [Prof. Gagandeep Singh](#)
- 2016–2020 **B.Tech. in Computer Science**, *Indian Institute of Technology Guwahati*, India
GPA: 9.66/10.0 (Institute Rank 3)

Research Interests

I am broadly interested in formal methods and automated reasoning. My current research focuses on developing precise and efficient program analyses using adaptable abstract interpretation, gradient-guided optimization, and efficient nonlinear reasoning. I am also interested in SAT/SMT solving, mathematical optimization, and neuro-symbolic systems that combine learning with formal reasoning.

Publications

Conferences

- PLDI 2026 **Evolving Abstract Transformers for Gradient-Guided, Adaptable Abstract Interpretation**
Shaurya Gomber, Debangshu Banerjee, Gagandeep Singh
- ESOP 2026 **Efficient Ranking Function-Based Termination Analysis via Bidirectional Decompositional Search**
Yasmin Sarita, Avaljot Singh, Shaurya Gomber, Mahesh Viswanathan, Gagandeep Singh

Workshops & Posters

- VerifAI ICLR 2026 **Unified Operational Formalism for LLM-based Theorem-proving Systems**
Avaljot Singh*, Shaurya Gomber*, Yasmin Sarita, José Meseguer, Gagandeep Singh
- VerifAI ICLR 2025 **Neural Abstract Interpretation**
Shaurya Gomber, Gagandeep Singh

Thesis

- MS CS 2024, UIUC **Neural Abstract Interpretation: Leveraging neural networks for automated, efficient and differentiable abstract interpretation**
Shaurya Gomber
🏆 [David J. Kuck Outstanding MS Thesis Award](#)

Awards & Fellowships

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| 2024 | David J. Kuck Outstanding Master's Thesis Award, UIUC | Details |
| 2024 | Richard T. Cheng Endowed Fellowship, UIUC | Details |
| 2019 | Institute Merit Scholarship, IIT Guwahati | Details |
| 2016 | KVPY Government of India Scholarship | Details |

Work Experience

- Summer 2023 **Applied Scientist Intern**, *Automated Reasoning Group @ Amazon AWS*, Santa Clara, USA
- Worked on **Zelkova**, an AWS tool for SMT-based reasoning over access policies.
 - Applied **SMTO** to design efficient encodings for type-casting semantics (e.g., numeric comparisons over strings), solving ~30k previously-unsolved production queries at ~1 min/query.
 - Fixed bugs and improved the oracle I/O interface in **CVC5**'s SMTO solver.
 - Tech Stack: Java, Scala, Python, SMT Solvers (Z3, CVC5)
- 2020-2022 **Senior Member of Technical Staff**, *D.E. Shaw & Co.*, Hyderabad, India
- Extended the firm's low-latency trading system (processing terabytes of data daily) with on-demand computation features to optimize trader workflows.
 - Contributed to design discussions for core trading system components and mentored two SDE-1s.
 - Tech Stack: Java, C++, React, Git, Bash, Grafana, NumPy, Matplotlib
- Summer 2019 **Software Engineering Intern**, *D.E. Shaw & Co.*, Hyderabad, India
- Implemented a *type-safe low-latency API* in Java to read and write on the firm's database, achieving a 60x run-time improvement in production-critical scripts, leading to a Pre-Placement Offer.

Teaching Experience

- Spring '26 Teaching Assistant, CS477 Formal Software Development Methods, UIUC
- Fall '23 Teaching Assistant, CS421 Programming Languages & Compilers, UIUC
- Spring '23 Teaching Assistant, CS421 Programming Languages & Compilers, UIUC
- Fall '22 Teaching Assistant, CS225 Data Structures & Algorithms, UIUC

Academic Service

- Artifact
Evaluation
Committee
- PLDI '25, PLDI '24

Talks

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| Jun 2026 | Evolving Abstract Transformers for Gradient-Guided Abstract Interpretation | Slides |
| | Programming Languages Design and Implementation (PLDI) 2026, Boulder, CO | |
| Oct 2025 | LINC: A Neuro-symbolic Approach for Logical Reasoning | Slides |
| | PL/FM/SE PhD Qualification Exam, UIUC | |
| May 2025 | Multi-Network Relational Verification and Certifiable Training | Slides |
| | CS584 Embedded System Verification, Spring 2025, UIUC | |
| Apr 2024 | Neural Abstract Interpretation | Slides |
| | Formal Methods Seminar, Spring 2024, UIUC | |
| Nov 2023 | Verification and Certified Training of PINNs | Slides |
| | CS598 Scientific Machine Learning, Fall 2023, UIUC | |
| Nov 2023 | Satisfiability and Synthesis Modulo Oracles | Slides |
| | Formal Methods Seminar, Fall 2023, UIUC | |
| May 2023 | Neural Approximations of Abstract Transformers | Slides |
| | CS477 Formal Software Development Methods, Spring 2023, UIUC | |
| Mar 2023 | Synthesizing Abstract Transformers | Slides |
| | Formal Methods Seminar, Spring 2023, UIUC | |
| Nov 2022 | Monotonic Neural Networks | Slides |
| | CS521 Trustworthy AI Systems, Fall 2022, UIUC | |

Mentorship

- **Peer Mentor, SAATHI Counselling Club:** Mentored CSE freshers under the [Mentor-Mentee program](#) at IIT Guwahati.
- **Placement Lectures Coordinator, IITG:** Taught Data Structures & Algorithms and managed content and scheduling for placement preparation lectures.
- **Treasurer, CSEA (2019–20):** Managed fund allocation for events of the Computer Science and Engineering Association, IIT Guwahati.

Selected Distinctions

- **Microsoft Code.Fun.Do 2019:** National finalist, top 10 of 300+ teams; built a Blockchain-based Voting System.
- **Inter IIT Tech Meet 2018:** Represented IITG in the coding hackathon at IIT Bombay.
- **ACM ICPC 2018:** Qualified for India regionals; represented IITG at Amritapuri, Kerala.
- **KVPY 2015:** AIR 178 among 1.5M candidates (top 0.01%) in IISc Bangalore's national science aptitude exam.
- **IIT JEE Advanced 2016:** AIR 902 among 1.5M candidates (top 0.06%) in India's engineering entrance exam.
- **IIT JEE Mains 2016:** AIR 2323 among 1.5M candidates (top 0.15%) in India's engineering entrance prelims.